Quantitative Research

Part 2: Graphical Representations

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The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
▶	**2 – Graphical Representations**	**Empirical CDF, histogram, boxplot**
	3 – Statistical Inference	Confidence intervals, hypothesis tests, p-values, one-sided tests, types of error
	4 – Regression	Correlation, simple and multiple linear regression, inference for regression, diagnostics
	5 – Classification	Logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers

All analyses are carried out in **R** (with RStudio); every deck comes with an R script that reproduces its numbers.

Contents – Part 2

1. Empirical Cumulative Distribution Function
2. Histogram
3. Boxplot
4. Exercise
5. R Commands
6. Summary

Learning objectives – Part 2

After this part you will be able to ...

- construct the **empirical CDF** and read off quantiles from it,
- build a **histogram** with equal and unequal class widths (densities!),
- construct a **boxplot** step by step, including the whiskers and outliers,
- **interpret** the shape of a distribution (symmetry, skewness, outliers) from these graphics,
- combine the numbers of Part 1 with the pictures of Part 2,
- create every graphic in R.

Key takeaway

The running story continues: the **shoe sizes of the soccer team** from Part 1 – now drawn instead of computed.

List of Symbols (1)

Symbol	Meaning	Symbol	Meaning
ω	Statistical unit	h_j	Absolute frequency
n	Number of statistical units	f_j	Relative frequency
X	Variable (characteristic)	H_j	Cumulative absolute frequency
x	Value of the variable	$F(x)$	Empirical CDF
x_i	Observed value	a_j	Class midpoint (class j)
$x_{(i)}$	i-th value of the ordered sample	$\tilde{x}$	Median
$\bar{x}$	Arithmetic mean	$\bar{x}_{\text{geo}}$	Geometric mean

List of Symbols (2)

Symbol	Meaning	Symbol	Meaning
s^2	Sample variance	v	Coefficient of variation
s	Standard deviation	v^*	Normalised coefficient of variation
R	Range	Δx_j	Class width (class j)
IQR	Interquartile range	$f_j/\Delta x_j$	Density (histogram)
$x_{0.25}$, $x_{0.75}$	25 % and 75 % quantile	k	Whisker factor (here 1.5)
Q_1, Q_3	1st and 3rd quartile	x_α	α-quantile, $F(x_\alpha) = \alpha$

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Empirical CDF – Construction

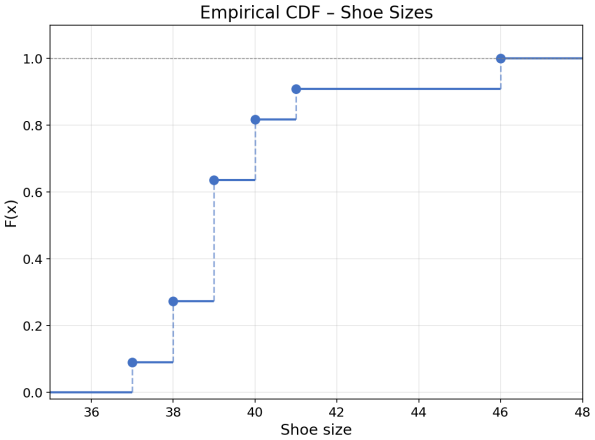
Construction: The observed values are plotted to scale on the x-axis; the corresponding y-values are given by the cumulative relative frequencies. The result is a step function whose y-values lie between 0 and 1.

- The height of each step depends on how often the value was observed
- Higher frequencies $\rightarrow$ taller steps

Frequency Table

Shoe size	37	38	39	40	41	46	Total
Absolute frequency	1	2	4	2	1	1	11
Relative frequency	0.0909	0.1818	0.3636	0.1818	0.0909	0.0909	1
Relative frequency in %	9.1 %	18.2 %	36.4 %	18.2 %	9.1 %	9.1 %	100 %
Relative frequency	$\frac{1}{11}$	$\frac{2}{11}$	$\frac{4}{11}$	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	1
Cumulative frequency	0.0909	0.2727	0.6364	0.8182	0.9091	1	
Cumulative frequency	$\frac{1}{11}$	$\frac{3}{11}$	$\frac{7}{11}$	$\frac{9}{11}$	$\frac{10}{11}$	1	

Empirical CDF



Empirical CDF – Definition

Definition

$$F(x) = \sum_{x_i \leq x} f(x_i)$$

- Shows the cumulative proportion of the sorted values
- Computing the quantiles for every observed value yields the empirical CDF
- Quantiles can therefore be read off directly from it

Empirical CDF – Reading Off Quantiles

Example: Median. The median is the smallest value at which the empirical CDF reaches 0.5:

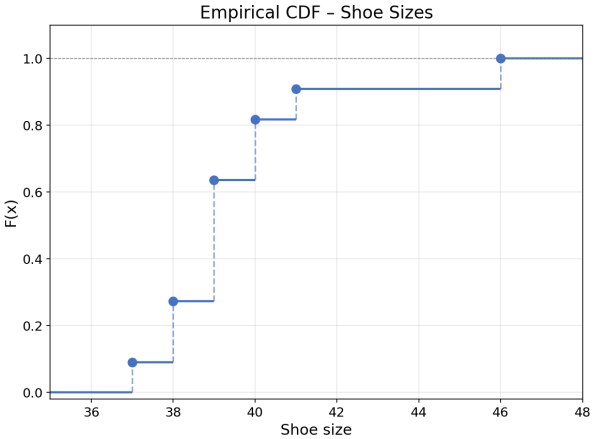
$$F(x_{0.5}) = 0.5 \quad \Leftrightarrow \quad F^{-1}(0.5) = x_{0.5}$$

Generalisation

Every quantile ($0 \leq \alpha \leq 1$) can be obtained from the empirical CDF:

$$F(x_\alpha) = \alpha \quad \Leftrightarrow \quad F^{-1}(\alpha) = x_\alpha$$

Empirical CDF



Empirical CDF in R

Computing

```
Fn.fussball ← ecdf(fussball)
```

Evaluating

```
Fn.fussball(39)    # value of the empirical CDF at x = 39
```

Plotting

```
plot(ecdf(fussball),
     verticals = TRUE,
     do.points = FALSE)
```

Overview

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Histogram

The histogram is a special type of bar chart used to visualise the distribution of metric data. What makes it special is that the observed values are not displayed individually but are grouped into **classes (bins/intervals)**.

Density

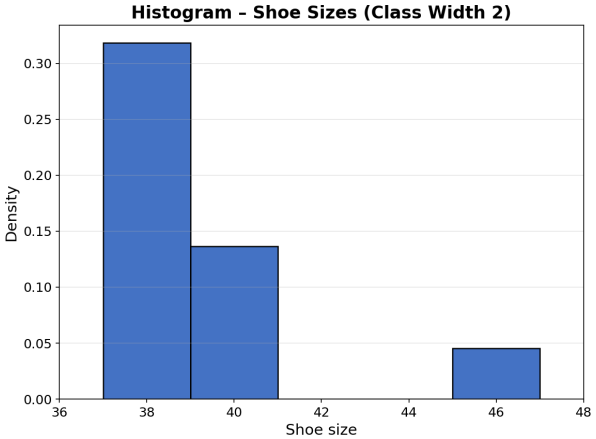
For each class, the **relative frequency** is computed and divided by the class width:

$$\text{density} = \frac{\text{relative frequency of the class}}{\text{class width}}$$

The bar height equals this density, so that the **area of each bar** represents the proportion of observations in the corresponding class.

- Classes are defined by interval boundaries
- The choice of class width has a strong effect on the shape of the histogram
- Classes of different widths are also possible

Histogram – Example



Histogram – Construction (Steps)

1. **Define the classes**
   - Notation: [= inclusive, (= exclusive
     - $[37, 39]$: both 37 and 39 belong to the class
     - $(39, 41]$: 39 does **not** belong, 41 does
   - Each class is open on the left and closed on the right: $(a, b]$
   - This way every value is assigned to **exactly one** class
   - Exception: the **first** class is closed on both sides $[a, b]$ so that the smallest value is not lost
   - e. g. $[37, 39]$, $(39, 41]$, $(41, 43]$, $(43, 45]$, $(45, 47]$
2. **Count the observations in each class** $\rightarrow$ absolute frequencies
3. **Compute the relative frequencies**
4. **Divide the relative frequencies by the class width** = density
5. **Draw the bars** with heights equal to these densities

Histogram – Construction (Table)

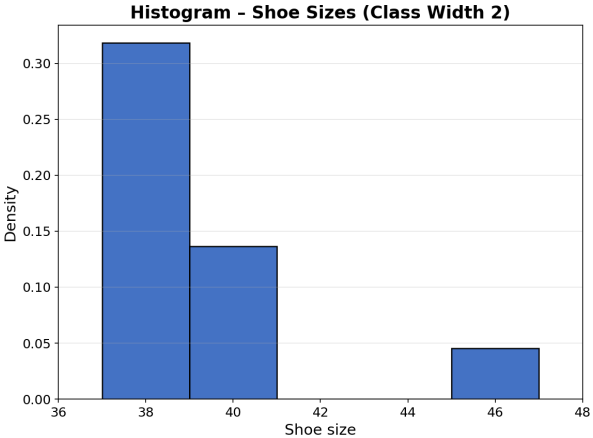
Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 39]$	$(39, 41]$	$(41, 43]$	$(43, 45]$	$(45, 47]$	Total
Class width	2	2	2	2	2	
Abs. frequency	7	3	0	0	1	11
Rel. frequency	0.6364	0.2727	0	0	0.0909	1
Density $= \frac{f_j}{\text{CW}}$	0.3182	0.1364	0	0	0.0455	

Example: first class

Density $= \frac{0.6364}{2} = 0.3182$

Histogram – Example



Histogram – Alternative Classes (Class Width 3)

Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 40]$	$(40, 43]$	$(43, 46]$	Total
Class width	3	3	3	
Abs. frequency	9	1	1	11
Rel. frequency	0.8182	0.0909	0.0909	1
Density $= \frac{f_j}{\text{CW}}$	0.2727	0.0303	0.0303	

Example: first class

Density $= \frac{0.8182}{3} = 0.2727$

Histogram – Unequal Class Widths

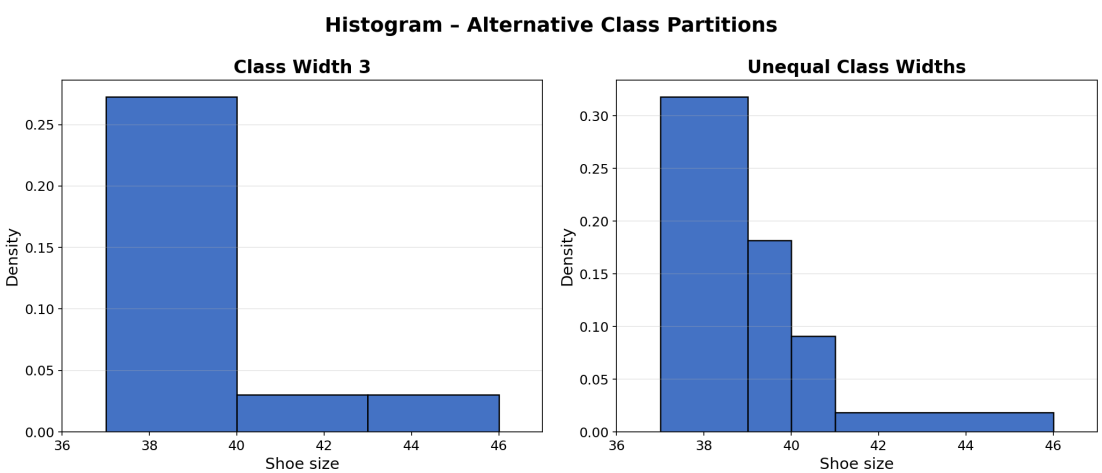
Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 39]$	$(39, 40]$	$(40, 41]$	$(41, 46]$	Total
Class width	2	1	1	5	
Abs. frequency	7	2	1	1	11
Rel. frequency	0.6364	0.1818	0.0909	0.0909	1
Density $= \frac{f_j}{\text{CW}}$	0.3182	0.1818	0.0909	0.0182	

Example: last class

Density $= \frac{0.0909}{5} = 0.0182$

Histogram – Alternative Class Partition



Histograms in R

Class width 2

```
hist(fussball, breaks = c(37,39,41,43,45,47), prob = TRUE)
hist(fussball, breaks = seq(37,47,2), prob = TRUE)  # alternative
```

Unequal class widths

```
hist(fussball, breaks = c(37,39,40,41,46), prob = TRUE)
```

Exercise – Histogram

Create a histogram for the soccer team's shoe sizes using a class width of 2.

The Data

Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39

Histogram – Construction (Table)

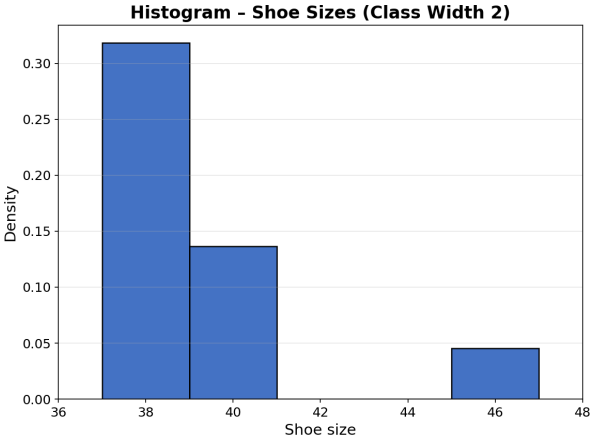
Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 39]$	$(39, 41]$	$(41, 43]$	$(43, 45]$	$(45, 47]$	Total
Class width	2	2	2	2	2	
Abs. frequency	7	3	0	0	1	11
Rel. frequency	0.6364	0.2727	0	0	0.0909	1
Density $= \frac{f_j}{\text{CW}}$	0.3182	0.1364	0	0	0.0455	

Example: first class

Density $= \frac{0.6364}{2} = 0.3182$

Histogram – Example



R Commands: Histogram

Histogram with equal class widths (here: 2)

```
hist(fussball, breaks=c(37,39,41,43,45,47), prob=TRUE)
hist(fussball, breaks=seq(37,47,2), prob=TRUE)
```

Histogram with unequal class widths

```
hist(fussball, breaks=c(37,39,40,41,46), prob=TRUE)
```

`prob=TRUE` plots the densities $f_j/\Delta x_j$ instead of the absolute frequencies. The `breaks` argument specifies the class boundaries.

R Commands: Classified Frequencies

Class partition

```
cut(fussball, breaks=c(value1,value2,...,valuen), include.lowest=TRUE)
```

Absolute classified frequencies

```
table(cut(fussball, breaks=c(value1,...,valuen), include.lowest=TRUE))
```

Relative classified frequencies

```
prop.table(table(cut(fussball, breaks=c(value1,...,valuen), include.lowest=TRUE)))
```

Overview

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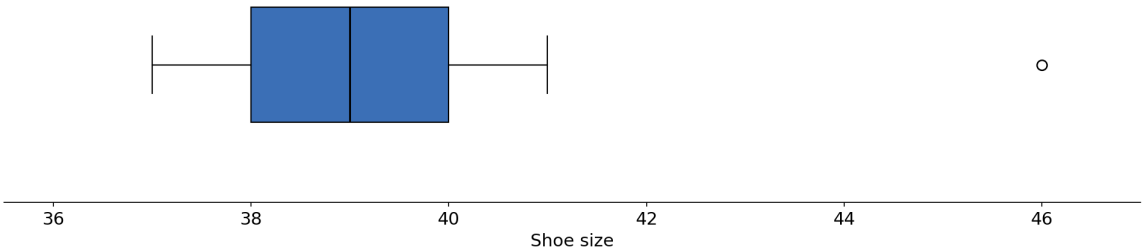
6 Summary

Boxplot

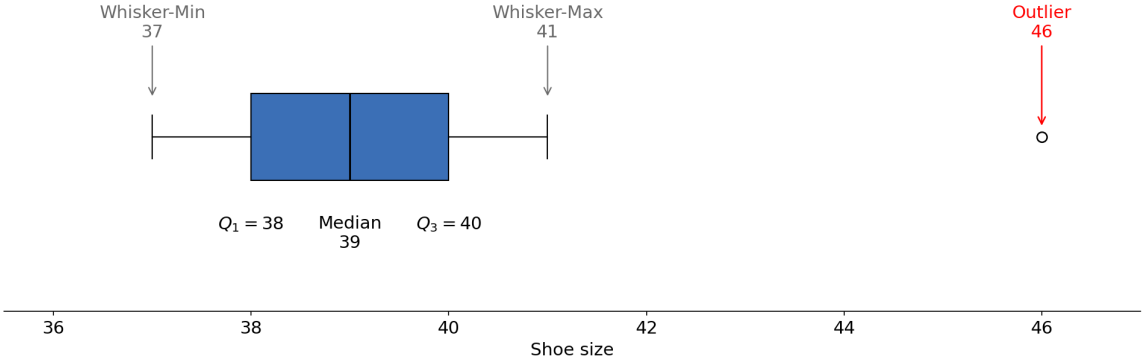
Definition

A graphical display that highlights only the most important summary values of a dataset and thereby visualises the distribution of the data.

Boxplot



Boxplot – with Labels



Boxplot – Construction

1. Sort the data
2. Determine the median
3. Determine the 1st quartile
4. Determine the 3rd quartile
5. Determine the lower whisker
6. Determine the upper whisker
7. Are there any outliers?
8. Draw the boxplot

Boxplot – Construction (Data)

Player	Shoe size	Sorted	
1	40	37	Lower whisker
2	39	38	
3	38	38	1st quartile
4	41	39	
5	46	39	
6	39	39	Median
7	39	39	
8	38	40	
9	37	40	3rd quartile
10	40	41	Upper whisker
11	39	46	Outlier
Sum	**436**	**436**	

Boxplot – Construction (Quartiles)

$n = 11$

	p	$n \cdot p$	$\lceil n \cdot p \rceil$ = position	Value
1st quartile	0.25	2.75	3	38
2nd quartile (median)	0.50	5.50	6	39
3rd quartile	0.75	8.25	9	40

Lower whisker	37
Q_1	38
Median	39
Q_3	40
Upper whisker	41
Outlier	46

Lower Whisker – Computation

The lower whisker is the **smallest data point that is not yet an outlier**. To find it, compute a lower bound – every value below it is classified as an outlier.

$$
\begin{aligned}
\text{Whisker}_{\text{min}} &= \min\{x_i \mid x_i \geq x_{0.25} - k(x_{0.75} - x_{0.25})\} \\
&\quad \text{with } k = 1.5 \\
&= \min\{x_i \mid x_i \geq 38 - 1.5(40 - 38)\} \\
&= \min\{x_i \mid x_i \geq 38 - 3\} \\
&= \min\{x_i \mid x_i \geq 35\} = 37
\end{aligned}
$$

Since 37 is the smallest data value ≥ 35, the lower whisker ends at 37.

Lower Whisker – Step by Step

1. **Compute the IQR:** the interquartile range is the difference between the 3rd and 1st quartile: $\text{IQR} = x_{0.75} - x_{0.25} = 40 - 38 = 2$
2. **Compute the lower bound:** subtract 1.5 times the IQR from the 1st quartile: $38 - 1.5 \cdot 2 = 38 - 3 = 35$
3. **Check the data:** every data point **below** 35 would be an outlier.
4. **Determine the lower whisker:** the smallest data value still ≥ 35 becomes the lower whisker. In our dataset this is 37.

Interpretation

The lower whisker extends down to 37 – the smallest value that is not classified as an outlier. If a value < 35 existed, it would be shown separately as an outlier in the boxplot.

Upper Whisker – Computation

The upper whisker is the **largest data point that is not yet an outlier**. To find it, compute an upper bound – every value above it is classified as an outlier.

$$\begin{aligned}\text{Whisker}_{\max} &= \max\{x_i \mid x_i \leq x_{0.75} + k(x_{0.75} - x_{0.25})\} \\ &\quad \text{with } k = 1.5 \\ &= \max\{x_i \mid x_i \leq 40 + 1.5(40 - 38)\} \\ &= \max\{x_i \mid x_i \leq 43\} = 41\end{aligned}$$

Since $46 > 43$, the value 46 is an outlier!

Upper Whisker – Step by Step

1. **Compute the IQR:** the interquartile range is the difference between the 3rd and 1st quartile: $\text{IQR} = x_{0.75} - x_{0.25} = 40 - 38 = 2$
2. **Compute the upper bound:** add 1.5 times the IQR to the 3rd quartile: $40 + 1.5 \cdot 2 = 40 + 3 = 43$
3. **Check the data:** every data point **above** 43 would be an outlier.
4. **Determine the upper whisker:** the largest data value still ≤ 43 becomes the upper whisker. In our dataset this is 41.

Interpretation

The upper whisker extends up to 41 – the largest value that is not classified as an outlier. Since $46 > 43$, the value 46 lies above the upper bound and is shown as a separate point in the boxplot.

Boxplot – Interpretation (1/3): Symmetry and Median

Symmetry

A boxplot is **symmetric** when the median lies in the centre of the box and both whiskers have the same length. This indicates that the data are evenly distributed around the centre.

Position of the median

The median splits the box into two halves. If it does **not** lie in the middle, the distribution is **skewed**:

- Median in the **left** half of the box $\rightarrow$ **right-skewed** (positively skewed): many small values, a few large values pull the distribution to the right.
- Median in the **right** half of the box $\rightarrow$ **left-skewed** (negatively skewed): many large values, a few small values pull the distribution to the left.

Boxplot – Interpretation (2/3): Box and Whiskers

Length of the box (IQR)

The box contains the middle 50 % of the data – from the 1st quartile (Q_1) to the 3rd quartile (Q_3). A **wide box** indicates that these middle values are widely spread. A **narrow box** shows that half of the data lie close together.

Length of the whiskers

The whiskers extend from the box to the smallest and largest value that is not an outlier. If both whiskers are **equally long**, the distribution is symmetric in the tails. **Different whisker lengths** indicate that the data are more spread out in one direction than in the other.

Boxplot – Interpretation (3/3): Outliers

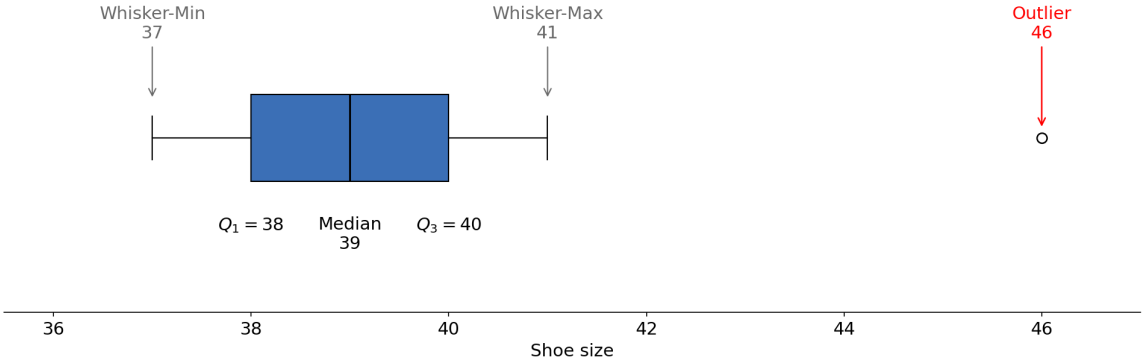
Outliers

Values that lie beyond the whiskers are shown individually as **outliers**. A value is considered an outlier if it lies more than 1.5 times the IQR below Q_1 or above Q_3. Outliers can indicate measurement errors, special cases, or a strongly skewed distribution.

Example: shoe-size data

- Box: $Q_1 = 38$, median $= 39$, $Q_3 = 40$ $\rightarrow$ the median lies in the centre, the box is symmetric.
- Whiskers: min $= 37$, max $= 41$ $\rightarrow$ both whiskers have the same length (each 1 unit from the box).
- Outlier: $46 > 43$ (upper bound) $\rightarrow$ despite the symmetric box, the overall boxplot is **not** symmetric.

Boxplot – with Labels



Boxplot – Conclusion

Advantages of the boxplot

- Compact display of the **most important summary statistics** at a glance: median, quartiles, range and outliers.
- Allows a quick assessment of the **symmetry** and **skewness** of a distribution.
- Especially useful for **comparing several groups** side by side – e. g. shoe sizes of different teams.
- **Outliers** are automatically detected and made visible.

Limitation

The boxplot reveals no information about the exact shape of the distribution inside the box – two very different datasets can produce the same boxplot. To study the shape, a histogram is a useful complement.

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Exercise: Body Heights – Problem Statement

The body heights (in cm) of $n = 15$ persons are studied:

164, 162, 168, 198, 166, 166, 192, 160, 151, 175, 180, 160, 168, 175, 168

Sorted:

151, 160, 160, 162, 164, 166, 166, 168, 168, 168, 175, 175, 180, 192, 198

1. Compute the mean, median, quartiles and the 17 % quantile.
2. Compute the four measures of dispersion.
3. Draw the empirical CDF, a histogram (class width 10 cm) and a boxplot.
4. Draw the empirical CDF for the classified data (class midpoints).

Exercise: Body Heights – The Data

Person	Height	Sorted	Abs. freq.	Rel. freq.	Cum. freq.
1	164	151	1	$0.0\overline{6}$	$0.0\overline{6}$
2	162	160	2	$0.1\overline{3}$	0.2
3	168	160			
4	198	162	1	$0.0\overline{6}$	$0.2\overline{6}$
5	166	164	1	$0.0\overline{6}$	$0.3\overline{3}$
6	166	166	2	$0.1\overline{3}$	$0.4\overline{6}$
7	192	166			
8	160	168	3	0.2	$0.\overline{6}$
9	151	168			
10	175	168			
11	180	175	2	$0.1\overline{3}$	0.8
12	160	175			
13	168	180	1	$0.0\overline{6}$	$0.8\overline{6}$
14	175	192	1	$0.0\overline{6}$	$0.9\overline{3}$
15	168	198	1	$0.0\overline{6}$	1
Sum	**2553**	**2553**	**15**	**1**	

Note: $\overline{6}$ denotes a repeating decimal, e. g.

$$0.0\overline{6} = 0.0666\ldots = \tfrac{1}{15}$$

Exercise: Body Heights – Results

Statistic	Value	Bound
Sum	2553	
Arithmetic mean	170.2	
Median	168	
Mode	168	
1st quartile	162	
3rd quartile	175	
17 % quantile	160	
Lower whisker	151	142.5
Upper whisker	192	194.5
Sample variance	151.60	
Standard deviation	12.3126	
Coefficient of variation	0.0723	
Normalised coeff. of variation	0.0187	

Exercise: Body Heights – Mean (Step by Step)

Formula

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Step 1: Sum all values:

$$\begin{aligned} \sum x_i &= 164 + 162 + 168 + 198 + 166 + 166 + 192 \\ &\quad + 160 + 151 + 175 + 180 + 160 + 168 + 175 + 168 \\ &= 2553 \end{aligned}$$

Step 2: Divide by $n = 15$:

$$\bar{x} = \frac{2553}{15} = 170.2\,\text{cm}$$

Exercise: Body Heights – Median and Mode

Sorted data:

151, 160, 160, 162, 164, 166, 166, **168**, 168, 168, 175, 175, 180, 192, 198

Median ($p = 0.5$)

$n \cdot p = 15 \cdot 0.5 = 7.5$ – not an integer $\rightarrow$ round up to position 8
The 8th value in the sorted sequence is $\tilde{x}_{0.5} = \mathbf{168}$ cm.

Mode

The most frequent value: 168 occurs 3 times $\rightarrow$ **168** cm.

Exercise: Body Heights – Quartiles and Quantile

Rule: Compute $n \cdot p$. If the result is **not an integer**, round up – that position gives the desired value.

17 % quantile ($p = 0.17$)

$n \cdot p = 15 \cdot 0.17 = 2.55 \rightarrow$ round up to position $3 \rightarrow x_{0.17} = 160\,\text{cm}$

1st quartile ($p = 0.25$)

$n \cdot p = 15 \cdot 0.25 = 3.75 \rightarrow$ round up to position $4 \rightarrow x_{0.25} = 162\,\text{cm}$

3rd quartile ($p = 0.75$)

$n \cdot p = 15 \cdot 0.75 = 11.25 \rightarrow$ round up to position $12 \rightarrow x_{0.75} = 175\,\text{cm}$

Exercise: Body Heights – Range and IQR

Range

$R = x_{\text{max}} - x_{\text{min}} = 198 - 151 = \mathbf{47}\,\text{cm}$

Interquartile range (IQR)

$\text{IQR} = x_{0.75} - x_{0.25} = 175 - 162 = \mathbf{13}\,\text{cm}$

The range reacts strongly to outliers (here: 198), whereas the IQR is unaffected by them because it only considers the middle 50 % of the data.

Exercise: Body Heights – Variance and Standard Deviation

Sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Step 1: Square the deviations from the mean:

$$(151 - 170.2)^2 = 368.64, \quad (160 - 170.2)^2 = 104.04, \quad \ldots$$

Step 2: Sum of squared deviations: $\sum(x_i - \bar{x})^2 = 2122.40$

Step 3: Divide by $n - 1 = 14$: $s^2 = \frac{2122.40}{14} = \mathbf{151.60}\,\text{cm}^2$

Standard deviation

$s = \sqrt{s^2} = \sqrt{151.60} = \mathbf{12.3126}\,\text{cm}$

Exercise: Body Heights – Coefficient of Variation

Coefficient of variation

$$v = \frac{s}{\bar{x}} = \frac{12.3126}{170.2} = \mathbf{0.0723}$$

The coefficient of variation expresses the dispersion relative to the mean – here the values scatter by about 7.2 % of the mean.

Normalised coefficient of variation

$$v^* = \frac{v}{\sqrt{n}} = \frac{0.0723}{\sqrt{15}} = \frac{0.0723}{3.8730} = \mathbf{0.0187}$$

The normalised coefficient of variation additionally takes the sample size into account – for larger n, v^* becomes smaller because more observations stabilise the estimate of the mean.

Exercise: Body Heights – Whisker Computation

Lower bound

$x_{0.25} - 1.5 \cdot \text{IQR} = 162 - 1.5 \cdot 13 = 162 - 19.5 = \mathbf{142.5}$

Smallest value ≥ 142.5: lower whisker $= 151$

Upper bound

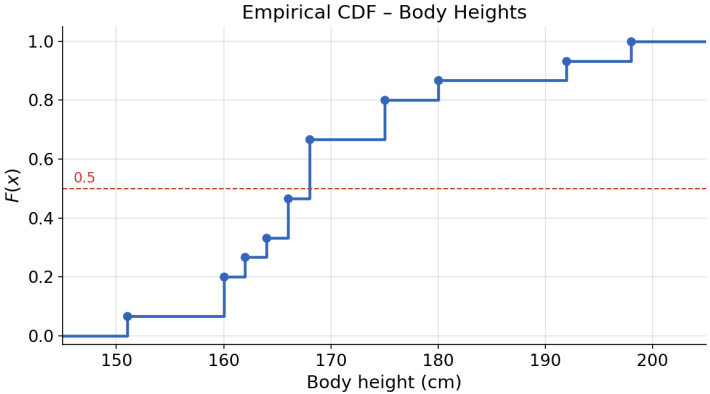
$x_{0.75} + 1.5 \cdot \text{IQR} = 175 + 1.5 \cdot 13 = 175 + 19.5 = \mathbf{194.5}$

Largest value ≤ 194.5: upper whisker $= 192$

Outlier

$198 > 194.5 \rightarrow$ the value 198 lies above the upper bound and is therefore an outlier.

Exercise: Body Heights – Empirical CDF



- Jump heights = relative frequencies (e. g. 0.2 at 168 cm, which occurs 3 times).
- The step function first reaches 0.5 at $x = 168 \Rightarrow$ median $\tilde{x} = 168$ cm.
- Values from the cumulative column of the data table.

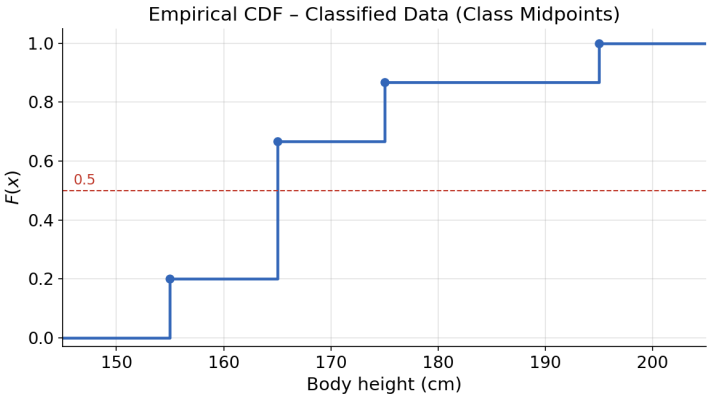
Exercise: Body Heights – Classified Data (Class Width 10 cm)

Class	$[150, 160]$	$(160, 170]$	$(170, 180]$	$(180, 190]$	$(190, 200]$	Total
Class midpoint a_j	155	165	175	185	195	
Abs. frequency h_j	3	7	3	0	2	15
Rel. frequency f_j	0.2	$0.4\overline{6}$	0.2	0	$0.1\overline{3}$	1
Cum. frequency $F(a_j)$	0.2	$0.\overline{6}$	$0.8\overline{6}$	$0.8\overline{6}$	1	

Construction

All observations of a class are placed at its **class midpoint** a_j; the empirical CDF then jumps by f_j at each midpoint.

Exercise: Body Heights – Empirical CDF of the Classified Data



- Only 4 steps instead of 10 – the classified CDF is much **coarser**.
- It reaches 0.5 at $a_2 = 165$, whereas the exact median is 168 cm.

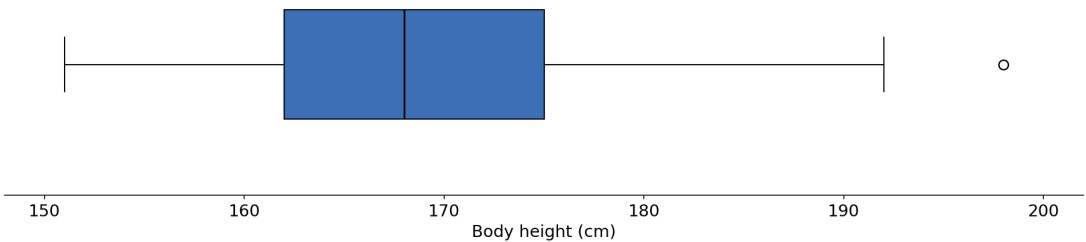
Interpretation

Classifying the data simplifies the picture but **loses information**.

Exercise: Body Heights – Histogram (Class Width 10 cm)



Exercise: Body Heights – Boxplot



Overview

1 Empirical Cumulative Distribution Function

2 Histogram

3 Boxplot

4 Exercise

5 R Commands

6 Summary

R Commands: Frequencies and Measures of Central Tendency

Absolute frequencies

```
tab.fussball ← table(fussball)
(tab.fussball ← table(fussball))  # assign and print
```

Mean, median and quantiles

```
mean(fussball)    # arithmetic mean
median(fussball)  # note: not our definition!
# Better: quantile (type=1 matches our definition):
quantile(fussball, 0.8, type=1)              # 80% quantile
quantile(fussball, c(0.25,0.5,0.75), type=1)  # quartiles
```

R Commands: Measures of Dispersion

Sample variance and standard deviation

```
var(fussball)
sd(fussball)
```

Coefficient of variation

```
(v ← sd(fussball)/mean(fussball))
```

Normalised coefficient of variation

```
(vstar ← v/sqrt(length(fussball)))
```

R Commands: Empirical CDF

Computing and evaluating

```
Fn.fussball ← ecdf(fussball)
Fn.fussball(39)  # value of the ECDF at x = 39
```

Plotting

```
plot(ecdf(fussball), verticals=TRUE, do.points=FALSE)
```

R Commands: Histogram

Equal class widths (here: 2)

```
hist(fussball, breaks=c(37,39,41,43,45,47), prob=TRUE)
hist(fussball, breaks=seq(37,47,2), prob=TRUE)
```

Unequal class widths

```
hist(fussball, breaks=c(37,39,40,41,46), prob=TRUE)
```

R Commands: Classified Frequencies

Absolute frequencies

```
table(fussball)
```

Class partition

```
cut(fussball, breaks=c(value1,value2,...,valuen), include.lowest=TRUE)
```

Absolute and relative classified frequencies

```
table(cut(fussball, breaks=c(value1,...,valuen), include.lowest=TRUE))
prop.table(table(cut(fussball, breaks=c(value1,...,valuen), include.lowest=TRUE)))
```

R Commands: Boxplot

Standard boxplot (Tukey)

```
boxplot(fussball, horizontal=TRUE, range=1.5)
```

range can be omitted for the default $k = 1.5$.

Boxplot with our quartile definition (type=1)

```
library(qboxplot)
qboxplot(fussball ~ 1, qtype=1, horizontal=TRUE)
```

Overview

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Summary (1/3) – Empirical CDF

Definition

$$F(x) = \sum_{x_i \leq x} f(x_i) \qquad \text{with values in } [0, 1]$$

Reading off quantiles

$$F(x_\alpha) = \alpha \quad \Longleftrightarrow \quad F^{-1}(\alpha) = x_\alpha, \qquad 0 \leq \alpha \leq 1$$

Special case: $F^{-1}(0.5) = \tilde{x}$ (median).

- Step function showing the cumulative relative frequency.
- Jump height at x_i equals $f(x_i)$.
- Every quantile can be read off directly.

Summary (2/3) – Histogram

Density of a class

$$\text{density}_j \;=\; \frac{f_j}{\Delta x_j} \;=\; \frac{\text{rel. frequency of class } j}{\text{class width}}$$

Area = relative frequency

$$\text{area}_j \;=\; \text{density}_j \cdot \Delta x_j \;=\; f_j, \qquad \sum_j f_j = 1$$

- Suitable for metric, classified data.
- Classes: $[a_1, a_2]$, $(a_2, a_3]$, $\ldots$, $(a_{k-1}, a_k]$ (first class closed on both sides).
- The choice of class width strongly affects the shape.
- Unequal class widths are also possible – bar heights remain densities.

Summary (3/3) – Boxplot

Five-number summary

$$x_{\text{min}} \text{ (or lower whisker)},\ x_{0.25},\ \tilde{x},\ x_{0.75},\ x_{\text{max}} \text{ (or upper whisker)}$$

$$\text{IQR} = x_{0.75} - x_{0.25}$$

Whisker bounds (Tukey, $k = 1.5$)

$$\text{Whisker}_{\text{min}} = \min\{x_i \mid x_i \geq x_{0.25} - k \cdot \text{IQR}\}$$
$$\text{Whisker}_{\text{max}} = \max\{x_i \mid x_i \leq x_{0.75} + k \cdot \text{IQR}\}$$

Outliers

All values $< x_{0.25} - 1.5 \cdot \text{IQR}$ or $> x_{0.75} + 1.5 \cdot \text{IQR}$ are displayed individually as outliers.

Outlook: Part 3 – Statistical Inference

What we have so far

Parts 1 and 2 **describe** the data at hand – in numbers and in pictures.

Next part

We treat the data as a **sample** and infer back to the **population**: confidence intervals and hypothesis tests. The mean $\bar{x}$ and the standard deviation s of Part 1 become the building blocks of every interval and every test.

Key takeaway

From describing a sample to drawing conclusions about the population – honestly reporting the uncertainty.